Aditi Sarker

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Research Interests

My research focuses on Multimodal Large Language Models (MLLMs), with an emphasis on hallucination detection and mitigation, multimodal alignment, spurious correlation removal, and robust representation learning. I develop methods to improve the reliability and grounding of vision-language models through representation-level interventions, optimization, and reinforcement learning with verifiable rewards. My broader research spans trustworthy multimodal AI, scalable learning, and robust and generalizable vision-language systems.

Education

Wayne State University **Aug. 2023 – Present(3.97 out of 4.00)**
PhD in Computer Science *Detroit, MI, USA*

Ph.D. Candidate (Expected 2028)

- Qualifying Exam (Survey Report) Title: The Role of Contrastive Learning in Multimodal Large Language Models (MLLMs).[Report]

Jahangirnagar University **Apr. 2017 – Nov. 2018(4.00 out of 4.00)**
MSc. in Computer Science and Engineering *Dhaka, Bangladesh*

- Thesis Title: Simulation of hazy image and validation of haze removal technique.

Jahangirnagar University **Dec. 2011 – Dec.2016(3.78 out of 4.00)**
BSc.in Computer Science and Engineering *Dhaka, Bangladesh*

Work Experiences

Graduate Teaching Assistant **August 2026 – Present**
Wayne State University *Detroit, Michigan*

- Currently serving as the laboratory instructor for the Software Engineering course and as the grader for the Introduction to Mobile Computing course.

Graduate Teaching Assistant **May 2026 – August 2026**
Wayne State University *Detroit, Michigan*

- Served as a lab instructor for the Senior Capstone Project course and supervised two student project teams in the development of mobile applications.

Graduate Teaching Assistant **August 2025 – May 2026**
Wayne State University *Detroit, Michigan*

- Served as a lab instructor for the Problem Solving and Programming course and the Object-Oriented Programming course.

Graduate Research Assistant **May 2025 – August 2025**
Wayne State University *Detroit, Michigan*

- Conducted research on multimodal large language models, focusing on hallucination mitigation, spurious correlation removal, and robust representation learning through large-scale experimentation.

Graduate Research Assistant **May 2024 – August 2024**
Wayne State University *Detroit, Michigan*

- Developed an algorithm on Hybrid Federated Learning and experiment.
- Developed algorithm on Federated Compositional Optimization and experiment.

Graduate Teaching Assistant **August 2023 – May 2024**
Wayne State University *Detroit, Michigan*

- Course: Problem Solving and Programming.
- Took lab class, tough programming in c++ language, grading, consultation, exam taking.

Assistant Professor **December 2022 – Auguest 2023**
Comilla University *Cumilla Bangladesh*

- Course: Data Mining, Artificial Intelligences, Algorithm Analysis and Design, Theory of Computation , Software Engineering.
- Took classes, Took lab classes, grading, consultation, exam taking.

Lecturer **December 2019 – December 2022**
Comilla University *Cumilla, Bangladesh*

- Course: Graph Theory, Object Oriented Programming, Deep Learning.
- Took classes, Took lab classes, grading, consultation, exam taking.

Fellowships and Awards

Thomas C. Rumble Fellowship **August 2024 – Present**
Wayne State University

WiCV @ CVPR 2026 Travel Grant **June 2026**
Women in Computer Vision (WiCV), CVPR 2026 *Denver, Colorado, USA*

Ongoing Research

Reinforcement Learning with Verifiable Rewards for Vision-Language Models **Oct. 2026 – Present**

- Investigating reinforcement learning with verifiable rewards (RLVR) for vision-language models, where automatically verifiable feedback is used to improve visual reasoning, perception, and response reliability without relying solely on learned reward models or human preference annotations.

Publications

Manuscripts Under Review

- **Sarker, A.**, Patchigolla, SP. V.,Sultan, R. I., Zhu, D., & Khanduri, P. *LIFE: LOCAL INTRINSIC DIMENSION-GUIDED FEATURE EDITING FOR ONE-TOKEN, ONE-LAYER HALLUCINATION REPAIR IN VISION-LANGUAGE MODELS*. Under review at **ICLR 2027**.

- **Sarker, A.**, Sultan, R. I., Zhu, H., Zhu, D., & Khanduri, P. *Hallucination Mitigation for Large Vision Language Models via Implicit Feature Stabilization*. Under review at **AAAI 2027**.

Conference Proceedings

- **Sarker, A.**, Shah, N., Sultan, R. I., Jang, R., Zhu, D., & Khanduri, P. *Partition-Aware Unlearning for Removing Spurious Correlations in Large Vision-Language Models*. Accepted at **NeurIPS 2026**. [Paper]

- Chandra Das, S., **Sarker, A.**, Saha, S., & Chakraborty, P. (2022). Covid-19 in bangladesh: An exploratory data analysis and prediction of neurological syndrome using machine learning algorithms based on comorbidity. In V. Skala, T. P. Singh, T. Choudhury,R. Tomar, M. Abul Bashar (Eds.), Machine intelligence and data science applications (pp. 595–608).doi:10.1007/978-981-19-2347-0

- Uddin, M. S., Gautam, B., **Sarker, A.**, Akter, M., & Haque, M. R. (2017). Image-based automated haze removal using dark channel prior. In 2017 ieee region 10 humanitarian technology conference (r10-htc) (pp. 412–415).doi:10.1109/R10-HTC.2017.828

- Feroz, M. A., Sultana, M., Hasan, M. R., **Sarker, A.**, Chakraborty, P., & Choudhury, T.(2022). Object detection and classification from a real-time video using ssd and yolo models. In A. K. Das, J. Nayak, B. Naik, S. Dutta, D. Pelusi (Eds.), Computational intelligence in pattern recognition (pp. 37–47). Singapore: Springer Singapore.

- Chakraborty, P., Roy, S., Sumaiya, S. N., & **Sarker, A.** (2023). Handwritten character recognition from image using cnn. In Micro-electronics and telecommunication engineering: proceedings of 6th ICMETE 2022 (pp. 37-47). Singapore: Springer Nature Singapore.

- Chakraborty, P., Yousuf, M. A., Islam, S., Khatun, M., **Sarker, A.**, & Rahman, S. (2022). Analysis of Texture Feature Extraction Technique in Image Processing. In Distributed Sensing and Intelligent Systems: Proceedings of ICDSIS 2020 (pp. 651-665). Cham: Springer International Publishing.

Journal Articles

- Khanduri, P., Li, C., Sultan, R. I., **Sarker, A.**, Qiang, Y., Kliewer, J., & Zhu, D. *On Federated Compositional Optimization: Algorithms, Analysis, and Guarantees.* **Transactions on Machine Learning Research (TMLR)**, 2026.
- **Sarker, A.**, Chakraborty, P., Sha, S. S., Khatun, M., Hasan, M. R., & Banerjee, K. (2020). Improvised technique for analyzing data and detecting terrorist attack using machine learning approach based on twitter data. Journal of Computer and Communications, 8 (7),50–62. doi:10.4236/jcc.2020.87
- **Sarker, A.**, Akter, M., & Uddin, M. S. (2019). Simulation of hazy image and validation of haze removal technique. Journal of Computer and Communications, 7 (2), 62–72. doi:10.4236/jcc.2019.72005

Workshop Publications

- **Sarker, A.**, & Khanduri, P. *PURGE: Partition-Aware Unlearning for Removing Spurious Correlations in Large Vision-Language Models.* Accepted at the **Women in Machine Learning (WiML) Workshop, NeurIPS 2026**. **Presented as a poster.** [Paper]
- **Sarker, A.**, & Khanduri, P. *Hallucination Mitigation for Large Vision Language Models via Implicit Feature Stabilization and Hierarchical Alignment Optimization.* Accepted at the **Women in Computer Vision (WiCV) Workshop, CVPR 2026**. [Paper]

Books and Chapters

- Uddin, M. S., Shorif, S. B., & **Sarker, A.** (2021). Role-framework of artificial intelligence in combating the covid-19 pandemic (M. A. R. Ahad A. Inoue, Eds.). doi:10.1007/978-3-030-75490-7

Extended Abstract

- **Sarkar, A.**, Rahman, M., & Uddin, M. S. (2020). Automated detection of plan diseases using r- cnn algorithms. IEEE CSBC Winter Symposium.
- Sheikh, R., **Sarkar, A.**, Uddin, M. S. (2020). Simulation of haze on real world images with generative adversarial network. IEEE CSBC Winter Symposium.

Projects(PhD)

Attention-Enhanced GraphSAGE with Skip-Fusion | *Python, C++* **January-April 2025**

- This project develops an attention-enhanced GraphSAGE model with skip-fusion that adaptively weights sampled neighbors and preserves node-specific features, improving robustness and representation quality in inductive graph learning while maintaining scalability.

What Can Transformers Learn In-Context? | *Python, C++* **January-April 2024**

- The project demonstrate that Transformers can be trained to master more intricate function classes, such as sparse linear functions, decision trees, and two-layer neural networks, achieving performance levels that match or surpass those of task-specific learning algorithms.

CARLA Self-Driving Motion Planning in Autonomous Driving | *Python, C++* **January-April 2024**

- We employ a functional motion planning stack. This is accomplished using behavioral planning logic, path selection, velocity profile development, and static collision checking. This project develops a functional motion planning stack for self-driving cars that allows them to avoid both static and dynamic barriers while tracking the center line of a lane and handling stop signs.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL
Developer Tools: VS Code, Eclipse, Google Cloud Platform, Android Studio
Technologies/Frameworks: Linux, Jenkins, GitHub, JUnit, WordPress

Professional Service

- Reviewer, *IEEE Signal Processing Letters*, 2026.

Certified Courses

What Is Generative AI? **September 2025**
LinkedIn Learning *Certificate of Completion*

Advance Machine Learning and Data Science **2020**
AILABS, HR Venture *Duration: 6 month*

Leadership / Extracurricular

CSE Society **August 2020 – August 2023**
Member *Comilla University*

Exam Committee **August 2020 – August 2023**
Member *Comilla University*

IWCI Workshop **2016**
Organiser *Jahangirnagar University*

References

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Wayne State University, MI
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